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QUESTION BANK

Semester – III Examinatation – 2026

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MAJ-4

MACHINE LEARNING

FUNDAMENTALS OF PYTHON

UNIT I – Fundamentals of Computers

2 Marks

1. Define a Computer and mention any two characteristics.
2. What is a Computer?
3. Mention the basic organization of a digital computer.
4. What are the different types of computers?
5. What is a Number System?
6. What is Binary Number System?
7. What is Decimal Number System?
8. What is ASCII?
9. What is Unicode?
10. What is a Computer Code?
11. Differentiate between Machine Level Language and High-Level Language.
12. What is an Assembler?
13. What is a Compiler?
14. What is an Interpreter?
15. What is Boolean Algebra?
16. What is a Truth Table?
17. What is an Algorithm?
18. What is a Flowchart?
19. What is Pseudocode?
20. What is a Translator Program?

7 Marks

1. Explain the evolution of computers with a brief description of different generations of computers.
2. Convert \((1101.101)_2\) into decimal and \((345)_{10}\) into binary.
3. Define Boolean Algebra and construct the truth table for the Boolean expression: A+(B.C).
4. Differentiate between Assembler, Compiler and Interpreter with examples.
5. Explain the basic organization of a digital computer with a diagram.
6. Explain different types of Number Systems with examples.
7. Explain different types of Computer Codes: BCD, ASCII, EBCDIC and Unicode.
8. Explain the characteristics and applications of computers.
9. Explain Algorithm and Flowchart with examples.
10. Explain the steps involved in planning a computer program.

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UNIT II – Python Basics

2 Marks

1. Mention two key features of Python Programming Language.
2. What is Python?
3. What are the different versions of Python?
4. What is Python IDLE?
5. What is an Identifier?
6. What are Keywords?
7. What are Variables?
8. What are Data Types?
9. What is Type Conversion?
10. What are Operators?
11. What are Expressions?
12. What is Indentation?
13. Explain the importance of Comments in Python.
14. Write the syntax for single-line and multi-line comments.
15. What is the use of type() function?
16. What is the use of id() function?
17. What are Python Libraries?
18. What is Console Input and Output?
19. What is a Frozen Set?
20. What is the difference between pop() and remove()?
21. What are Membership Operators?

7 Marks

1. Explain the different features of Python Programming Language.
2. Explain the use of Type Conversion Functions in Python.
3. Explain the difference between Identity and Membership Operators.
4. Describe the difference between Lists and Tuples.
5. Explain Python Variables, Data Types and Operators with examples.
6. Explain Identifiers, Keywords and Statements in Python.
7. Explain the different types of Operators in Python with examples.
8. Explain Python Libraries and how to import libraries with examples.
9. Explain Console Input and Console Output in Python.
10. Explain the installation and execution of a simple Python program.

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UNIT III – Python Control Flow

2 Marks

1. What is Control Flow?
2. What is an if statement?
3. What is an if-else statement?
4. What is an elif statement?
5. What is a while loop?
6. What is a for loop?
7. What is the use of break statement?
8. What is the use of continue statement?
9. What is the use of range() function?
10. What is the use of exit() function?
11. Differentiate between break and continue.
12. What is a Nested Loop?
13. What is an Illustrative Program?

7 Marks

1. Explain the if-elif-else statement in Python with an example program.
2. Explain the role of loops in Python with an example.
3. Write a Python program to check whether a number is prime or not.
4. Explain different types of Control Flow Statements with examples.
5. Explain the for loop and while loop with examples.
6. Explain the use of break and continue statements with examples.
7. Explain the range() function with suitable examples.
8. Write a Python program to find the factorial of a number.
9. Write a Python program to check whether a number is even or odd.
10. Write a Python program to find the largest among three numbers.

UNIT IV – Python Functions and Strings

2 Marks

1. What is a Function in Python?
2. What are the different types of Functions?
3. What is Function Definition?
4. What is Function Calling?
5. What are Parameters and Arguments?
6. What is a Return Statement?
7. What are Default Parameters?
8. What are Command Line Arguments?
9. What are Keyword Arguments?
10. What are Arbitrary Arguments?
11. What is *args?
12. What is **kwargs?
13. What is a String in Python?
14. What is String Slicing?

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15. What is String Concatenation?
16. What is String Comparison?
17. What are Escape Sequences?
18. What is String Formatting?
19. What is Raw and Unicode String?
20. What is String Traversing?

7 Marks

1. Discuss the different types of Functions in Python and illustrate with an example.
2. Describe the role of Return Statement in Functions.
3. Explain the difference between *args and **kwargs.
4. Explain String Slicing and Concatenation with suitable examples.
5. What are Escape Sequences in Python? Explain with examples.
6. Explain Function Definition, Function Calling and Passing Parameters with examples.
7. Explain Default Parameters and Keyword Arguments with examples.
8. Explain different operations on Strings with examples.
9. Explain String Formatting and String Traversing.
10. Write a Python program to reverse a string.
11. Write a Python program to check whether a string is a palindrome.
12. Write a Python program to count the number of vowels in a string.

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